PROJECT REPORT

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

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1. INTRODUCTION

1.1 Project Overview

The project, “Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer,” is a ServiceNow-based automation solution for standard laptop procurement. The solution connects the Service Catalog, Requested Items, approval processing, Flow Designer, and Catalog Tasks to create a simple end-to-end workflow.

When a Standard Laptop request is approved, the configured Flow Designer automation creates a Catalog Task and assigns it to the Hardware team for configuration. The project is intended to remove manual task creation and manual assignment from the approved-request stage.

The solution provides a consistent transition from approval to hardware processing, improves task visibility, reduces avoidable delays, and supports prompt initiation of laptop configuration.

1.2 Purpose

The purpose of the project is to automate a repetitive IT procurement activity that would otherwise require a person to create and assign a task after approval. The automation ensures that an approved Standard Laptop request is converted into an actionable Catalog Task and routed to the Hardware team through a defined Flow Designer process.

- Reduce manual effort in Catalog Task creation.
- Automate assignment of the generated task to the Hardware team.
- Improve visibility of post-approval procurement work.
- Reduce delays and manual errors caused by repetitive task handling.
- Provide a consistent and traceable approval-to-task workflow.

2. IDEATION PHASE

2.1 Problem Statement

In the existing Standard Laptop procurement process, approval of a request does not by itself guarantee that the corresponding hardware configuration task is created and assigned immediately. Manual creation and assignment of a Catalog Task can introduce delays, inconsistent processing, and avoidable errors.

The project addresses this problem by configuring ServiceNow Flow Designer to detect an approved Standard Laptop request, create the required Catalog Task, and assign it to the Hardware team. This provides an automated and repeatable transition from approval to hardware processing.

Problem statement: How can the approved Standard Laptop procurement process be automated so that a Catalog Task is created and assigned to the Hardware team consistently, without manual task creation and assignment?

2.2 Empathy Map Canvas

The empathy map considers the people involved in the procurement workflow and the practical difficulties experienced when task creation and assignment depend on manual actions.

Perspective	Project-Relevant Understanding
Says	The laptop request has been approved; the Hardware team should be able to start configuration immediately.
Thinks	Approved requests should not remain waiting because a task has not been manually created or assigned.
Does	Submits a Standard Laptop request, follows the approval process, and expects the request to progress to hardware processing.
Feels	May experience uncertainty or frustration when there is a delay between approval and the start of hardware work.
Pain Points	Manual task creation, manual assignment, processing delays, inconsistent handling, and limited visibility after approval.
Needs / Gains	Automatic Catalog Task creation, Hardware team assignment, consistent processing, and better workflow visibility.

2.3 Brainstorming

The brainstorming phase focused on simple ServiceNow-native approaches that could remove the manual step between request approval and hardware processing.

Idea	Description	Priority
Flow Designer automation	Trigger the process from the approved Standard Laptop request and create the Catalog Task automatically.	High
Automatic Hardware assignment	Assign the generated Catalog Task to the Hardware team as part of the same flow.	High
Approval-to-task integration	Use the approval result as the control point for starting the procurement task workflow.	High
Task visibility improvement	Ensure the generated Catalog Task is visible to the team responsible for configuration.	Medium
Manual checklist	Use a manual checklist for task creation and assignment when automation is unavailable.	Low / Backup

The selected idea was Flow Designer automation because it directly addresses the two main manual activities: Catalog Task creation and Hardware team assignment.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

The customer journey follows the requester's experience from submitting a Standard Laptop request to the point at which the Hardware team can begin configuration.

Journey Stage	Requester / User Action	System / Process Response	Expected Experience
1. Request	Submit Standard Laptop request through Service Catalog.	ServiceNow records the request and Requested Item.	Request is captured clearly.
2. Approval	Request moves through the approval process.	Approval status is evaluated.	User waits for the defined approval decision.
3. Approved	Standard Laptop request receives approval.	Flow Designer automation starts.	No manual task creation is expected.

4. Task Creation	No separate requester action is required.	Flow creates a Catalog Task.	Work item is generated automatically.
5. Assignment	No separate requester action is required.	Catalog Task is assigned to Hardware team.	Correct team receives the work.
6. Hardware Processing	Hardware team begins laptop configuration.	Team works on the generated Catalog Task.	Approved request moves into actionable processing.

3.2 Solution Requirement

The solution requirements define the functional and operational behavior required for the project.

ID	Requirement	Description	Priority
FR-01	Approval Trigger	The automation shall respond to an approved Standard Laptop request.	High
FR-02	Catalog Task Creation	The automation shall create a Catalog Task for the approved request.	High
FR-03	Hardware Assignment	The generated Catalog Task shall be assigned to the Hardware team.	High
FR-04	Service Catalog Integration	The workflow shall use the Standard Laptop request and Requested Item context.	High
FR-05	Approval Integration	The approval state shall control the transition to automated task creation.	High
FR-06	End-to-End Continuity	The configured components shall work as one approval-to-task workflow.	High
FR-07	Task Visibility	The created task shall provide an actionable work item for the Hardware team.	Medium
FR-08	Consistent Processing	Approved Standard Laptop requests shall follow the same configured automation path.	High

3.3 Data Flow Diagram

The data flow represents how the Standard Laptop procurement information moves through ServiceNow components.

Level	Flow
Level 0	IT Procurement User → Service Catalog → Approval Process → Flow Designer → Catalog Task → Hardware Team
Level 1	Standard Laptop Request → Requested Item → Approval Status → Approved Request → Flow Designer → Create Catalog Task → Assign Hardware Team → Laptop Configuration

Primary data elements: Standard Laptop Request, Requested Item (RITM), Approval Status, Flow Designer Flow, Catalog Task, and Hardware Team assignment.

The DFD emphasizes that the approval result is the control point for automation and that the final operational output is a Catalog Task assigned to the Hardware team.

3.4 Technology Stack

Component / Technology	Role in the Project
ServiceNow Platform	Provides the overall environment for service procurement configuration and workflow automation.
Service Catalog	Provides the request entry point for the Standard Laptop procurement request.
Requested Item (RITM)	Represents the requested item context used by the procurement workflow.
Approval Process	Provides the approval status that controls the automated transition.
Flow Designer	Implements the automation logic for approved requests, Catalog Task creation, and assignment.
Catalog Task	Represents the actionable hardware-processing work item.
Hardware Team	Receives the generated task and performs laptop configuration.

The project is configuration- and workflow-oriented within ServiceNow. No external programming language, external database, or third-party service is required for the defined automation.

4. PROJECT DESIGN

4.1 Problem Solution Fit

Identified Problem	Project Solution	Fit / Value
Manual Catalog Task creation	Flow Designer automatically creates a Catalog Task after approval.	Directly removes the repetitive manual step.
Manual Hardware team assignment	Flow Designer assigns the generated task to the Hardware team.	Routes approved work consistently.
Delay after approval	Automation starts the next defined workflow step after approval.	Reduces avoidable waiting.

Manual processing errors	Repeatable configured flow handles task creation and assignment.	Improves consistency.
Disconnected approval and hardware processing	ServiceNow components are connected through Flow Designer.	Creates a continuous approval-to-task workflow.
Limited task visibility	Generated Catalog Task provides an actionable work item for Hardware.	Improves visibility of pending hardware work.

4.2 Proposed Solution

The proposed solution uses ServiceNow Flow Designer to automate the transition from an approved Standard Laptop request to hardware processing. The Standard Laptop request is submitted through the Service Catalog and proceeds through the approval process. When the request is approved, the configured flow creates a Catalog Task and assigns it to the Hardware team.

Step	Process
1	Requester submits a Standard Laptop request through Service Catalog.
2	The request is represented through the appropriate Requested Item context.
3	The request proceeds through the configured approval process.
4	Flow Designer evaluates the approved state and starts the automated path.
5	Flow Designer creates the required Catalog Task.
6	The Catalog Task is assigned to the Hardware team.
7	Hardware team can begin laptop configuration using the generated task.

The scope is intentionally limited to the defined procurement automation: approved Standard Laptop request → Flow Designer → Catalog Task creation → Hardware team assignment.

4.3 Solution Architecture

Architecture Layer	Components	Purpose
Request Layer	Service Catalog → Standard Laptop Request → Requested Item	Captures the procurement request and item context.
Approval Layer	Approval Process → Approval Status → Approved Request	Determines whether the request is ready for automated processing.
Automation Layer	Flow Designer → Evaluate approved request → Create Catalog Task → Assign Hardware Team	Performs the defined automation.
Task Layer	Catalog Task → Hardware Team → Laptop Configuration	Provides the actionable work item and operational destination.

End-to-end architecture flow: Service Catalog → Requested Item → Approval → Approved Request → Flow Designer → Catalog Task → Hardware Team → Laptop Configuration.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project plan organizes the implementation from requirement understanding through ServiceNow configuration, Flow Designer development, integration validation, testing, and documentation.

Phase	Planned Activities	Expected Output
1. Requirement Understanding	Review the procurement problem, approval behavior, and required automation.	Problem definition and requirements.
2. Solution Design	Map requirements to ServiceNow components and define the approval-to-task flow.	Solution design and architecture.
3. Service Catalog / Request Configuration	Verify the Standard Laptop request and Requested Item context used by the workflow.	Configured request context.
4. Flow Designer Development	Configure automation for approved requests, Catalog Task creation, and Hardware team assignment.	Configured automation flow.
5. Integration	Verify Service Catalog, Requested Item, approval, Flow Designer, and Catalog Task interaction.	Integrated procurement workflow.
6. Testing & Validation	Test approval-triggered task creation, Hardware assignment, visibility, and consistency.	Validated implementation.
7. Documentation	Prepare project report and supporting documentation.	Final project documentation.

Planned Activity Schedule

Activity	Dependency	Planned Sequence	Deliverable

Requirement identification	Project problem definition	1	Requirement understanding
Solution and architecture design	Requirements	2	Architecture / workflow design
Service Catalog / request configuration	Solution design	3	Request context
Flow Designer implementation	Request context	4	Automation flow
Integration / configuration review	Automation flow	5	Integrated workflow
Functional and performance testing	Configured workflow	6	Validation results
Final documentation	Testing and validation	7	Completed report

Roles and Responsibilities

Role	Responsibility
Project / Team Members	Understand requirements, coordinate implementation, review the complete workflow.
Configuration & Development	Configure ServiceNow request context and Flow Designer automation.
Testing & Validation	Execute test scenarios and verify Catalog Task creation and Hardware team assignment.
Documentation	Prepare project report, workflow details, testing records, and final submission.

Key Milestones

- Requirements and problem definition completed.
- Solution architecture and workflow identified.
- Standard Laptop request context verified.
- Flow Designer automation configured.
- Catalog Task creation and Hardware team assignment validated.
- Final documentation completed.

Risk and Mitigation Planning

Potential Risk	Mitigation
Incorrect approval condition	Verify that the automation evaluates the intended approved state before task creation.
Catalog Task not created	Test the flow with an approved Standard Laptop request and inspect the flow execution.
Incorrect Hardware team assignment	Verify the configured assignment value and validate the generated task.
Incomplete integration between components	Test the complete path from Service Catalog through approval and Flow Designer to Catalog Task.
Configuration and documentation mismatch	Review the implemented workflow against the documented requirements before final submission.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Performance testing for this configuration-based ServiceNow project focuses on operational responsiveness, consistency, reliability, and reduction of manual effort across the approval-to-task workflow. The project documentation does not define fixed response-time benchmarks; therefore, no unsupported numerical performance values are stated.

Test ID	Test Scenario	Expected Result	Status
PT-01	Approved Standard Laptop request triggers the configured Flow Designer automation.	Flow execution starts for the approved request.	Pass / Fail
PT-02	Flow processes an approved request and reaches Catalog Task creation.	A Catalog Task is created automatically.	Pass / Fail
PT-03	Generated Catalog Task is checked for Hardware team assignment.	Catalog Task is assigned to the Hardware team.	Pass / Fail
PT-04	Complete approval-to-task workflow is observed from start to finish.	Workflow completes without manual task creation or assignment.	Pass / Fail
PT-05	Multiple approved Standard Laptop requests are processed.	Each request follows the same defined task creation and assignment process.	Pass / Fail
PT-06	Workflow is checked for processing issues during normal usage.	No unnecessary workflow interruption is observed.	Pass / Fail

Performance Acceptance Criteria

- The Flow Designer automation is triggered for an approved Standard Laptop request.
- A Catalog Task is created automatically.
- The Catalog Task is assigned to the Hardware team.
- The workflow completes without requiring manual Catalog Task creation or assignment.
- Repeated approved requests follow the same configured workflow.
- Actual execution-time values, if required, are recorded during project demonstration or validation.

7. RESULTS

7.1 Output Screenshots

The project output is demonstrated through ServiceNow configuration and workflow screens showing the Standard Laptop request, approval state, Flow Designer automation, generated Catalog Task, and Hardware team assignment. Screenshots should be inserted on the reserved blank page included at the end of this report.

Expected Output	What the Screenshot Should Demonstrate
Standard Laptop Request	The relevant Service Catalog request / Requested Item context.
Approval	The request reaching the approved state.
Flow Designer	The configured automation that handles the approved request.
Catalog Task	Automatic creation of the task for the approved Standard Laptop request.
Hardware Assignment	The generated Catalog Task assigned to the Hardware team.
End-to-End Result	The completed transition from approved request to actionable hardware task.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Automates Catalog Task creation after Standard Laptop approval.
- Automatically assigns the generated task to the Hardware team.
- Reduces manual effort and repetitive task handling.
- Improves consistency in procurement processing.
- Reduces avoidable delays between approval and hardware configuration.
- Improves visibility of the actionable hardware work item.
- Uses ServiceNow Flow Designer for a maintainable workflow-based solution.
- Provides a simple end-to-end connection between request approval and hardware processing.

Disadvantages

- The solution depends on correct ServiceNow configuration of the request, approval, and flow.
- Incorrect approval conditions can prevent the automation from starting as intended.
- Incorrect assignment configuration can route the task to the wrong team.
- The current scope is focused on Standard Laptop procurement and does not automatically cover other hardware categories.
- Performance measurements require actual execution during testing; fixed response-time benchmarks are not defined in the project scope.
- Changes to ServiceNow configuration may require corresponding updates to the flow and documentation.

9. CONCLUSION

The Standard Laptop Procurement Automation project demonstrates how ServiceNow Flow Designer can streamline a repetitive IT procurement activity. The solution connects the Service Catalog, Requested Items, approval processing, Flow Designer, and Catalog Tasks into a defined workflow.

The key automation occurs after approval: Flow Designer creates a Catalog Task and assigns it to the Hardware team. This removes the need for manual task creation and assignment, improves process consistency, increases visibility, and helps the Hardware team begin laptop configuration promptly.

The project therefore provides a focused, practical automation for standard laptop procurement while maintaining a clear and manageable ServiceNow workflow.

10. FUTURE SCOPE

- Extend the same automation pattern to other standard IT hardware categories.
- Introduce additional routing rules when different hardware categories require different support teams.
- Add more detailed SLA-oriented monitoring for procurement and hardware tasks.
- Provide broader reporting and analytics on request volume, task completion, and processing delays.
- Introduce notification steps for relevant users or teams when task states change.
- Enhance exception handling for rejected, cancelled, or incomplete requests.
- Add reusable automation components for a broader IT procurement workflow.
- Integrate additional approval and fulfillment conditions if future procurement requirements demand them.

Future enhancements should remain aligned with the ServiceNow platform configuration and the organization's procurement policies.

11. APPENDIX

Source Code (if any)

Not applicable to the current scope. The solution is primarily implemented through ServiceNow Service Catalog, approval configuration, Flow Designer, and Catalog Task configuration rather than a standalone source-code application.

Dataset Link

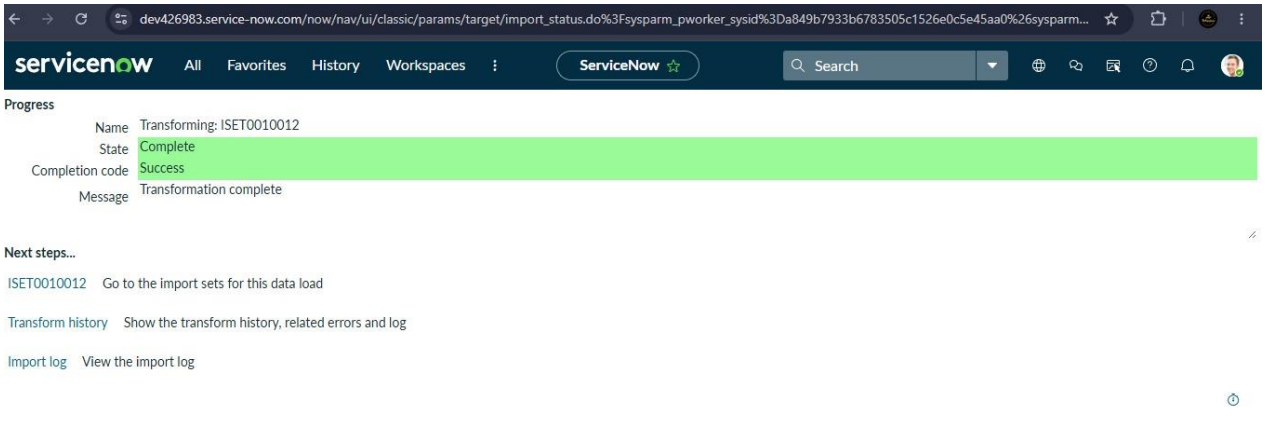
Not applicable. The project does not require an external dataset for the defined Standard Laptop procurement automation.

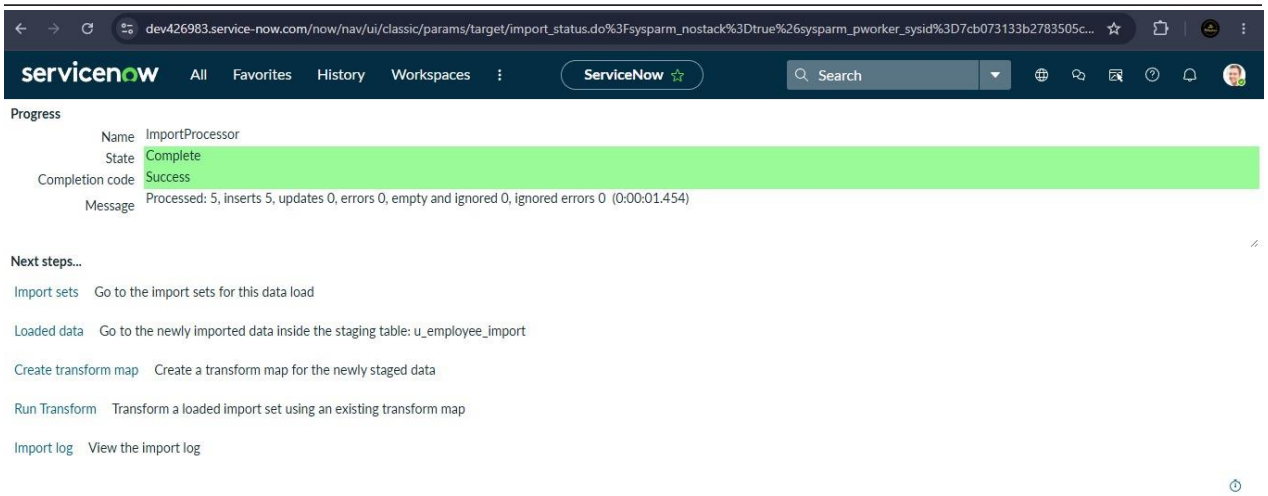
GitHub & Project Demo Link

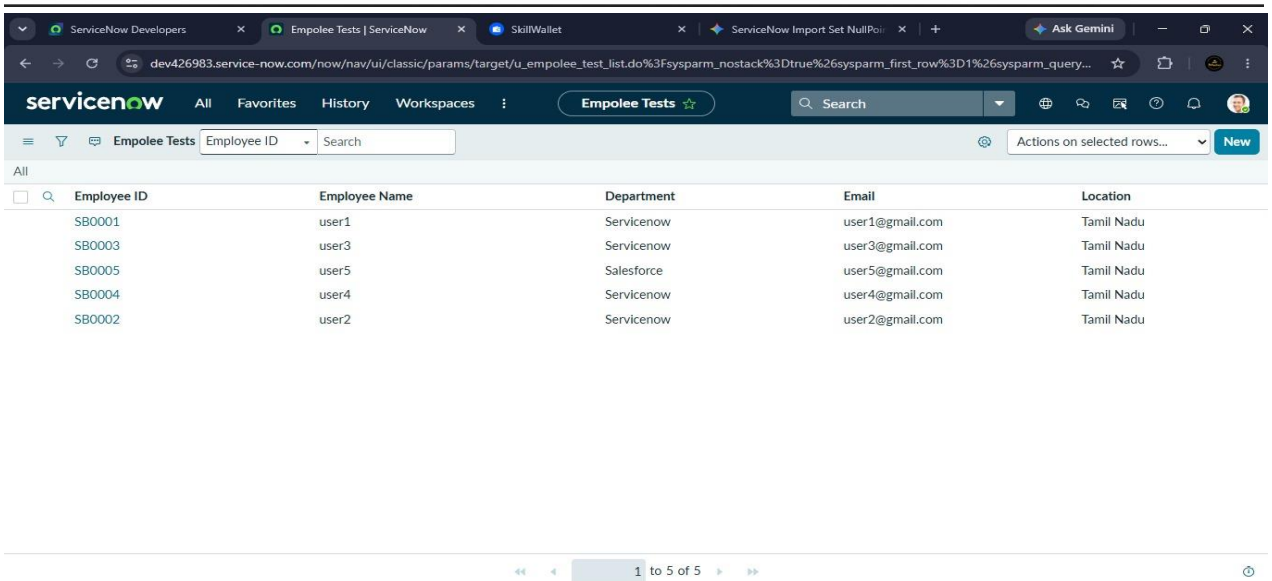
GitHub: Not applicable to the current configuration-based project scope. Project Demo Link: To be added after the project demonstration or institutional submission process, if a link is provided.

Project Identification

Field	Details
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Team ID	SWTID-2026-7837
Project Type	ServiceNow workflow automation
Primary Platform	ServiceNow
Core Automation	Approved Standard Laptop request → Flow Designer → Catalog Task → Hardware Team
Primary Outcome	Automated and consistent post-approval hardware task creation and assignment.


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servicenow
All
Favorites
History
Workspaces
ServiceNow
Search
Progress
Name Transforming: ISET0010012
State Complete
Completion code Success
Message Transformation complete
Next steps...
ISET0010012 Go to the import sets for this data load
Transform history Show the transform history, related errors and log
Import log View the import log


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servicenow
All
Favorites
History
Workspaces
ServiceNow
Search
Progress
Name ImportProcessor
State Complete
Completion code Success
Message Processed: 5, inserts 5, updates 0, errors 0, empty and ignored 0, ignored errors 0 (0:00:01.454)
Next steps...
Import sets Go to the import sets for this data load
Loaded data Go to the newly imported data inside the staging table: u_employee_import
Create transform map Create a transform map for the newly staged data
Run Transform Transform a loaded import set using an existing transform map
Import log View the import log


ServiceNow Developers
Empolee Tests | ServiceNow
SkillWallet
ServiceNow Import Set NullPoi
Ask Gemini
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Empolee Tests
Employee ID
Search
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Email
Location
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